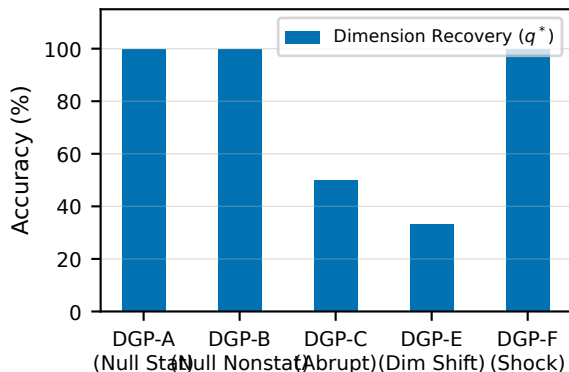
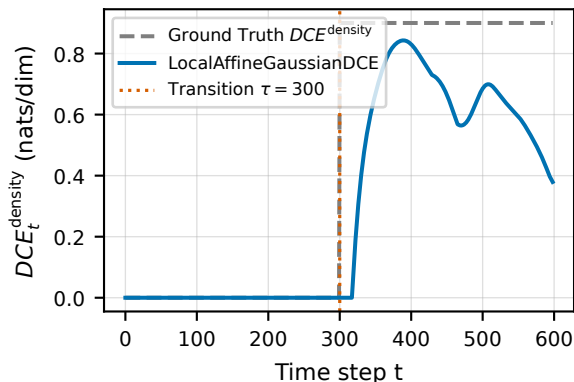
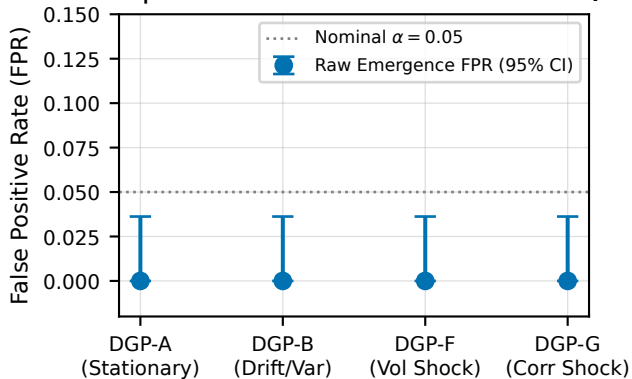


a | Abrupt Causal Emergence Tracking (DGP-b) | Dimension Recovery Accuracy $P(\hat{q}^* = q^*)$ **c | False Positive Rate under Nulls (R=100)****d | Tracking Error Across Benchmark DGPs**